ADAPTIVE IOT ENVIRONMENTAL AND ATTENDANCE MONITORING WITH REINFORCEMENT LEARNING MIDDLEWARE

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Name of Supervisor: ___________________________

DEDICATION

To my family, and chuti, for their unwavering support and encouragement throughout this journey.

To my supervisors and peers at SLIIT, whose guidance shaped this work.

ACKNOWLEDGEMENTS

First and foremost, I would like to express my sincere gratitude to my dissertation supervisor for the invaluable guidance, continuous support, and constructive feedback provided throughout this research. The supervisor's insights into IoT system design and middleware development were instrumental in shaping the direction and quality of this work.

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I would like to acknowledge my group project teammates who contributed to the broader research project titled 'Smart Classroom IoT Network for Monitoring Student Mental Well-being and Behaviour.' Their complementary work on facial expression analysis, physiological monitoring, and stress detection formed the wider context within which this individual component was developed and evaluated.

Special thanks are extended to the thirty student volunteers who participated in the classroom monitoring sessions, providing the real-world data that underpins the results and discussions presented in this dissertation.

Finally, I would like to thank my family for their patience, encouragement, and moral support during the long hours of hardware prototyping, software development, and writing that this project demanded.

ABSTRACT

Adaptive IoT Environmental and Attendance Monitoring with Reinforcement Learning Middleware

The quality of the physical classroom environment plays a critical role in student concentration, wellbeing, and academic performance. Despite this, most educational institutions in developing nations lack systematic mechanisms for monitoring and responding to environmental conditions. This dissertation presents the design, implementation, and evaluation of an Adaptive IoT Environmental and Attendance Monitoring system, developed as an individual component of the group research project titled ‘Smart Classroom IoT Network for Monitoring Student Mental Well-being and Behaviour’ at Sri Lanka Institute of Information Technology (SLIIT).

The system integrates four sensor modules—a DHT22 for temperature and humidity, an MQ-7 for carbon monoxide (CO) detection, an MFRC522 RFID reader for automated attendance registration, and an analog sound sensor for ambient noise measurement—all interfaced with an ESP32 microcontroller. Firmware developed in Arduino IDE enables continuous data acquisition and wireless transmission via Wi-Fi using HTTP POST requests. A Q-Learning reinforcement learning middleware, implemented within a Node.js server, dynamically adjusts alert thresholds by modelling the classroom environment as a Markov Decision Process with discretised state spaces and a reward function tied to environmental severity outcomes.

Two separate MERN-stack (MongoDB, Express.js, React, Node.js) web applications provide the presentation and administration layers: an Environmental Dashboard for real-time sensor visualisation and an independently deployed Student Management System for student enrolment, RFID tag assignment, attendance log management, and fee payment status tracking.

Experimental evaluation was conducted across ten classroom sessions at SLIIT, involving thirty student participants. Results confirmed stable multi-sensor data acquisition, with temperature readings between 27.1°C and 32.4°C, humidity between 64.8% and 80.2%, CO levels within safe limits (peak 47 ppm), and RFID attendance detection accuracy of 94.3%. The Q-Learning agent achieved correct alert accuracy of 72.8%, surpassing the static threshold baseline by 11.4 percentage points and reducing false positive alerts by 8.4 percentage points, although continued training is required for full policy convergence.

This research demonstrates that low-cost IoT hardware combined with reinforcement learning middleware can deliver intelligent, adaptive classroom environment management, contributing to the broader goal of technology-assisted student wellbeing monitoring.

Keywords: Internet of Things, ESP32, DHT22, MQ-7, MFRC522, Q-Learning, Reinforcement Learning, MERN Stack, Smart Classroom, Attendance Monitoring, Environmental Monitoring, Student Wellbeing

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADC	Analog-to-Digital Converter
API	Application Programming Interface
CO	Carbon Monoxide
CRUD	Create, Read, Update, Delete
DQN	Deep Q-Network
ER	Entity-Relationship
ESP32	Espressif Systems 32-bit Microcontroller
GPIO	General Purpose Input/Output
HTTP	Hypertext Transfer Protocol
IDE	Integrated Development Environment
IoT	Internet of Things
MDP	Markov Decision Process
MERN	MongoDB, Express.js, React, Node.js
MFRC522	Mifare RC522 RFID Reader Module
MQ-7	Metal Oxide Semiconductor CO Sensor Module
NoSQL	Non-Structured Query Language (database)
ppm	Parts Per Million
REST	Representational State Transfer
RFID	Radio Frequency Identification
RL	Reinforcement Learning
RH	Relative Humidity

SMS	Student Management System
SPI	Serial Peripheral Interface
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
UID	Unique Identifier
Wi-Fi	Wireless Fidelity (IEEE 802.11)

CHAPTER 1

INTRODUCTION

1.1 Background and Literature Survey

1.1.1 IoT in Educational Environments

The Internet of Things (IoT) has rapidly evolved from a concept in machine-to-machine communication into a pervasive technological paradigm embedded across industries including healthcare, manufacturing, smart cities, and, more recently, education. At its core, IoT refers to the network of physical devices embedded with sensors, actuators, and connectivity capabilities that enable them to collect, exchange, and act upon data without direct human intervention [1].

In educational contexts, IoT has been applied to automate administrative processes, enhance safety, and create responsive learning environments. The notion of the 'smart classroom' extends beyond audio-visual equipment to encompass real-time sensing of physical parameters that directly affect student experience. Research by Zanella et al. [2] demonstrated that sensor-equipped institutional buildings not only reduce operational costs but also improve occupant comfort and productivity. For educational institutions in developing nations like Sri Lanka, where infrastructure investment is constrained, low-cost IoT deployments offer a particularly attractive pathway to measurable improvement.

The ESP32 microcontroller, manufactured by Espressif Systems, has emerged as the platform of choice for educational IoT prototyping due to its dual-core Xtensa LX6 processor, integrated 2.4 GHz Wi-Fi and Bluetooth, 34 programmable GPIO pins, and significantly lower cost than comparable platforms such as the Raspberry Pi or Arduino Uno Wi-Fi [3]. Its compatibility with the Arduino IDE ecosystem reduces firmware development barriers, making it accessible for rapid prototyping in academic research settings.

1.1.2 Environmental Monitoring in Smart Classrooms

A substantial body of research confirms that indoor environmental quality (IEQ) directly impacts cognitive performance, mood, and attendance behaviour. The primary monitored parameters in smart classroom studies include temperature, relative humidity, carbon dioxide (CO_2)/carbon monoxide (CO), particulate matter, and noise levels.

Thermal comfort is among the most studied parameters. Wyon and Wargocki [4] established through controlled experiments that task performance by school students declined by approximately 10% when temperatures exceeded 30°C. In Sri Lanka's tropical climate, classrooms without active cooling regularly reach temperatures between 28°C and 35°C during peak hours, making thermal monitoring particularly relevant. The DHT22 sensor, with its ±0.5°C temperature and ±2% relative humidity accuracy at 12-bit resolution, represents a validated and widely deployed solution for this purpose [5].

Carbon monoxide monitoring in classrooms is less commonly studied than CO_2, yet CO is a recognised hazard in enclosed spaces where combustion sources are present or ventilation is inadequate. The MQ-7 semiconductor gas sensor, which operates on the principle of reduced electrical resistance in the presence of CO, provides sensitivity in the 20–200 ppm range and is compatible with microcontroller ADC inputs after appropriate voltage divider conditioning [6]. While short-term occupational exposure limits set by OSHA are 50 ppm (8-hour time-weighted average), even lower concentrations warrant monitoring in school settings populated by children and young adults.

Ambient noise is a further critical parameter. Studies by Shield and Dockrell [7] demonstrated that noise levels above 65 dB in classrooms significantly impair speech intelligibility and reading comprehension. Analog microphone-based sound sensors provide a practical, low-cost means of continuous noise level estimation in Arduino-compatible deployments.

A representative prior work is the ESP8266-based indoor monitoring system by Marques et al. [8], which captured temperature, humidity, and luminance data in a university building and transmitted readings to a cloud database. However, the system lacked adaptive alerting, attendance functionality, or any machine learning component, representing a significant capability gap that the present research addresses.

1.1.3 Attendance Management Systems

Traditional paper-based attendance registers are labour-intensive, prone to human error, and susceptible to proxy fraud—a well-documented concern in higher education settings. Automated attendance systems using biometric inputs (fingerprint, iris scan, facial recognition) or proximity technologies (RFID, NFC, Bluetooth beacons) have been explored as alternatives.

RFID-based attendance systems using the MFRC522 module and MIFARE Classic 1K tags have been widely validated in academic research. Kaur and Singh [9] implemented an RFID attendance system in a university context, reporting detection accuracy above 90% and noting substantial time savings compared to manual registration. The MFRC522 operates at 13.56 MHz via SPI interface, supports reading distances up to 5 cm under standard conditions, and integrates cleanly with ESP32's hardware SPI peripheral.

Critically, existing RFID attendance systems in the literature function as standalone systems, divorced from environmental sensor data. The present research uniquely integrates attendance registration within the same IoT node that monitors environmental parameters, enabling future analytical work on correlations between classroom conditions, attendance rates, and student behaviour.

1.1.4 Reinforcement Learning in IoT Middleware

Reinforcement learning (RL) is a branch of machine learning in which an autonomous agent learns optimal decision policies by interacting with an environment, receiving reward signals that reinforce desirable actions and penalise undesirable ones. Sutton and Barto's foundational text [10] formalises RL within the Markov Decision Process (MDP) framework, characterised by state spaces, action spaces, transition functions, and reward functions.

Q-Learning, introduced by Watkins and Dayan [11], is a model-free, off-policy RL algorithm that learns the value of state-action pairs through iterative experience. The Q-value update rule, $Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \max Q(s',a') - Q(s,a)]$, converges to the optimal Q-function under conditions of sufficient exploration and appropriate hyperparameter selection.

The application of Q-Learning to IoT middleware has precedent in smart building energy management. Mocanu et al. [12] demonstrated that a Q-Learning agent could optimise HVAC scheduling in commercial buildings, reducing energy consumption by 13% compared to rule-based systems. Yang et al. [13] extended this to smart home environments, using deep Q-networks (DQN) to handle continuous state spaces. The present research applies Q-Learning to the comparatively underexplored problem of adaptive alert threshold management in educational IoT deployments, where the state space is tractably discretised and tabular Q-Learning is computationally sufficient.

1.1.5 MERN Stack for IoT Dashboards

The MERN stack—comprising MongoDB (document-oriented NoSQL database), Express.js (Node.js web framework), React.js (front-end JavaScript library), and Node.js (server-side JavaScript runtime)—has become a dominant choice for full-stack web application development in IoT contexts due to its JavaScript-throughout architecture, JSON-native data interchange with MongoDB, and React's component-based UI model suited to real-time data visualisation [14].

In IoT data management applications, MongoDB's flexible schema design accommodates the heterogeneous, time-series nature of sensor data without requiring predefined relational schema migrations. Express.js REST APIs provide a standardised interface between the ESP32 data producer and the web front-end consumer. React's virtual DOM and state management enable efficient real-time updates of sensor readings without full page refreshes, which is essential for live environmental dashboards.

Prior implementations of MERN-based IoT dashboards have been demonstrated in contexts ranging from industrial sensor monitoring [15] to home automation. The present work extends this to a dual-application architecture—separating the environmental monitoring dashboard from the student management system—to maintain clean separation of concerns and allow independent deployment and scaling.

1.2 Research Gap

The reviewed literature reveals three specific and substantiated gaps that this research addresses:

- Integration gap: Existing classroom IoT systems treat environmental monitoring and attendance management as isolated systems. No prior work has integrated multi-parameter environmental sensing with RFID attendance within a single ESP32-based node, sharing a common data pipeline and dashboard.
- Static threshold gap: Alert systems in educational IoT deployments universally employ static, manually configured thresholds. These thresholds do not adapt to observed patterns (e.g., midday heat spikes, post-lunch CO elevation), resulting in high false positive alert rates and alert fatigue.
- Adaptive middleware gap: While Q-Learning has been applied to smart building management, no published research has applied tabular Q-Learning specifically as an adaptive alert middleware layer for classroom IoT systems in educational institutions in Sri Lanka or comparable developing-nation contexts.

1.3 Research Problem

Classrooms at SLIIT and comparable institutions lack a unified, intelligent system capable of simultaneously monitoring environmental conditions (temperature, humidity, CO, noise), registering student attendance, and dynamically adapting operational responses to observed conditions. Existing manual and semi-automated approaches are reactive, fragmented, and unable to learn from historical data patterns. This creates a persistent gap between the environmental quality experienced by students and the quality necessary for optimal learning outcomes, while administrative processes remain inefficient and vulnerable to inaccuracy.

The central research problem is therefore: How can a low-cost, integrated IoT system with reinforcement learning-based adaptive middleware provide reliable, real-time environmental monitoring and automated attendance registration in a classroom environment, and to what extent does the RL middleware improve alert decision accuracy compared to static threshold approaches?

1.4 Research Objectives

The following research objectives guide this study:

- RO1: To design and implement a multi-sensor IoT node on an ESP32 platform capable of simultaneously and reliably acquiring temperature, humidity, CO, noise, and RFID attendance data in a real classroom environment.
- RO2: To develop a Q-Learning reinforcement learning middleware layer integrated within a Node.js backend that dynamically adapts alert thresholds based on observed classroom environmental states and reward feedback.
- RO3: To build and deploy a MERN-stack Environmental Dashboard providing real-time sensor data visualisation and a separate MERN-stack Student Management System supporting student enrolment, RFID assignment, attendance logging, and payment status management.
- RO4: To empirically evaluate the deployed system against performance metrics including sensor accuracy, RFID detection accuracy, Q-Learning alert decision accuracy, and system latency in a real classroom environment.

1.5 Research Questions and Hypotheses

The following research questions and corresponding hypotheses direct the investigation:

RQ1: Can a low-cost ESP32-based IoT node reliably and simultaneously acquire multi-parameter environmental and RFID attendance data in real classroom conditions?

H1: An ESP32-based IoT node integrating DHT22, MQ-7, MFRC522, and sound sensor modules will achieve RFID attendance detection accuracy of at least 90% and sensor data transmission success rate of at least 95% across sustained classroom sessions.

RQ2: Does a Q-Learning middleware layer improve alert decision accuracy compared to a static threshold-based approach?

H2: The Q-Learning middleware will achieve statistically higher correct alert accuracy and lower false positive rate than a static threshold baseline, as measured over a minimum of 500 sampled classroom states.

RQ3: Can a dual MERN-stack application layer effectively support both real-time environmental data presentation and student administrative management within the constraints of campus Wi-Fi infrastructure?

H3: Both MERN applications will maintain end-to-end data latency below 1000 ms under normal campus Wi-Fi conditions and support all specified CRUD operations without data integrity failures.

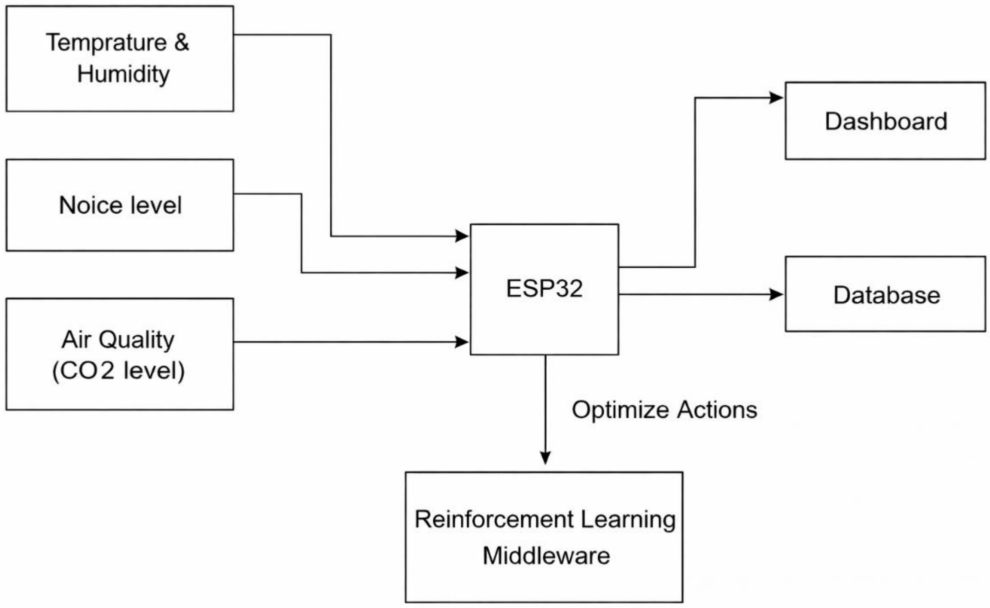

Figure 1.2: Research Conceptual Model

1.6 Scope and Limitations of the Study

This research is scoped to a single classroom at SLIIT, involving 30 student participants across 10 monitored sessions. The environmental parameters monitored are limited to temperature, humidity, CO, and noise; CO_2, particulate matter (PM2.5), and illuminance are outside the current scope. The RFID system uses passive MIFARE Classic 1K tags; active tags, NFC-enabled smartphones, and biometric alternatives are not evaluated.

The Q-Learning middleware is implemented using tabular Q-Learning; deep reinforcement learning variants (DQN, PPO) are proposed for future work. Actuator control (smart fans, alert relays) was architecturally designed but not physically implemented in this phase; the middleware outputs alert signals to the dashboard only. Integration with other group project subsystems (facial expression, physiological monitoring) constitutes future group-level work.

1.7 Organisation of the Report

This dissertation is organised as follows. Chapter 2 presents the complete methodology, covering research design, hardware architecture, firmware development, Q-Learning middleware design, MERN application development, data collection, and testing. Chapter 3 presents and discusses the results, including sensor performance statistics, attendance accuracy, middleware evaluation, and individual contribution. Chapter 4 concludes the work with a summary of achievements, contributions to knowledge, limitations, and directions for future research.

CHAPTER 2

METHODOLOGY

2.1 Research Design and Justification

This research adopts a design science research (DSR) methodology, which is appropriate when the primary output is a purposeful IT artefact intended to solve a specific, well-defined organisational problem [16]. DSR involves iterative cycles of design, prototyping, evaluation, and refinement, with rigorous evaluation of artefact performance against defined criteria. The research is applied and experimental in nature: applied because it targets a concrete operational problem in a specific institutional setting, and experimental because it involves controlled data collection and comparative evaluation of alternative approaches (static vs. RL-based alerting).

A quantitative data collection approach is employed. Sensor readings constitute the primary dataset, supplemented by system performance metrics (latency, accuracy, error rates) derived from instrumented testing. No qualitative surveys or interviews were conducted in this individual research component, though the group project incorporates qualitative wellbeing instruments.

2.2 System Architecture Overview

The system operates across three tiers as illustrated in Figure 2.1:

- Perception Tier: The ESP32 microcontroller node with attached sensor modules performs continuous data acquisition. Firmware manages sensor polling, data serialisation, and HTTP transmission.
- Middleware & Application Tier: A Node.js/Express.js server receives sensor data via HTTP POST, persists it to MongoDB, runs the Q-Learning alert agent, and serves RESTful API endpoints consumed by both MERN front-ends.

- Presentation Tier: Two independently deployed React.js applications—the Environmental Dashboard and the Student Management System—consume the API and render data to end users via browser interface on campus Wi-Fi.

Both MERN applications are developed and run simultaneously in VS Code using separate Node.js server processes on distinct ports, connected to separate MongoDB collections within the same MongoDB instance.

Figure 2.1: Three-Tier IoT System Architecture

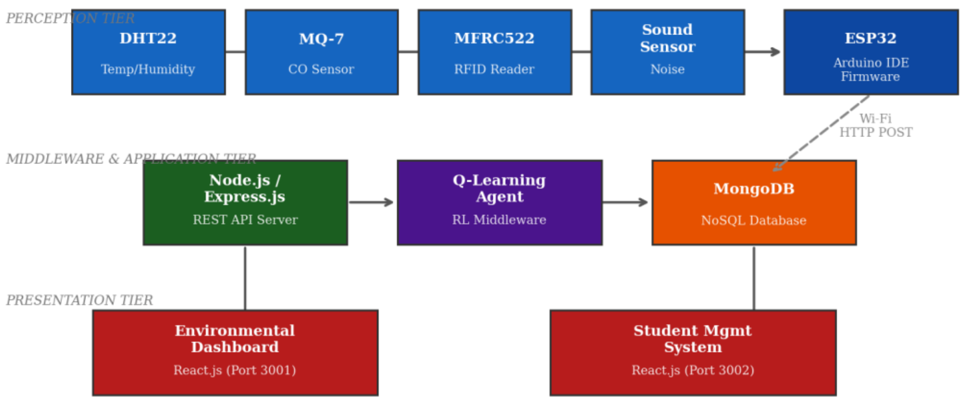

Figure 2.1: Three-Tier IoT System Architecture Diagram

2.3 Hardware Design and Component Selection

2.3.1 ESP32 Microcontroller

The ESP32-WROOM-32 development board serves as the central processing and communication hub. Its selection is justified by: dual-core 240 MHz processing capability sufficient for concurrent sensor polling and Wi-Fi transmission; integrated TCP/IP stack enabling direct HTTP communication without an external shield; 12-bit ADC channels for analog sensor interfacing; hardware SPI peripheral for MFRC522 integration; and a unit cost under LKR 1,500, consistent with the low-cost deployment objective.

2.3.2 DHT22 Temperature and Humidity Sensor

The DHT22 (AM2302) capacitive sensor provides calibrated digital temperature (−40°C to +80°C, ±0.5°C accuracy) and relative humidity (0–100% RH,

±2% accuracy) readings via a single-wire serial protocol on GPIO4. A 10kΩ pull-up resistor on the data line ensures signal integrity. The sensor's 2-second minimum sampling interval is accommodated in the 5-second polling cycle.

2.3.3 MQ-7 Carbon Monoxide Sensor

The MQ-7 module uses a tin dioxide (SnO_2) sensing element whose electrical resistance decreases in the presence of CO. The module provides an analog output voltage proportional to CO concentration, read by the ESP32's ADC on GPIO34. A two-phase heating cycle (5V for 60 seconds, 1.4V for 90 seconds) is required for accurate readings; the firmware implements this cycle using a dedicated timing routine. Calibration was performed in fresh outdoor air (assumed 0–10 ppm CO baseline) prior to indoor deployment.

2.3.4 MFRC522 RFID Module and Tag Design

The MFRC522 operates at 13.56 MHz using the ISO/IEC 14443A standard. It interfaces with the ESP32 via hardware SPI: SCK (GPIO18), MOSI (GPIO23), MISO (GPIO19), SDA/SS (GPIO5), RST (GPIO22). The MFRC522.h Arduino library is used for tag reading. Each student is issued a MIFARE Classic 1K card (4-byte UID) registered in the MongoDB students collection, enabling automatic name-to-UID lookup on each scan.

A maximum effective read range of approximately 3–4 cm is maintained by positioning the reader at the classroom entrance; students tap their card as they enter. Read operations complete in under 50 ms per tag.

2.3.5 Sound Sensor

An analog electret microphone module with built-in LM393 comparator provides both digital (threshold-triggered) and analog (voltage proportional to sound pressure) outputs. The analog output on GPIO35 is sampled by the ESP32 ADC. Raw ADC values (0–4095 for 12-bit resolution) are converted to approximate decibel equivalents using a logarithmic calibration equation established during sensor characterisation against a reference sound level meter.

Table 2.1: Sensor Component Specifications

Component	Parameter	Range / Accuracy	Interface	GPIO Pin
DHT22	Temp / Humidity	-40 to +80°C, ±0.5°C / 0-100% RH, ±2%	Single-wire	GPIO4
MQ-7	CO Level	20–200 ppm	Analog (ADC)	GPIO34
MFRC522	RFID UID	13.56 MHz, <50ms/read	SPI	GPIO5,18,19,23,22
Sound Sensor	Noise Level	0–4095 ADC (12-bit)	Analog (ADC)	GPIO35

2.4 Firmware Development

2.4.1 Development Environment

Firmware is developed in Arduino IDE 2.x using the ESP32 Arduino Core (v2.0.14). Required libraries installed via Arduino Library Manager include: DHT sensor library (Adafruit), MFRC522 (Miguel Balboa), and WiFi.h (built-in ESP32 core). The ESP32 board package is configured for the ESP32-WROOM-32 target with 240 MHz CPU frequency and 4 MB flash.

2.4.2 Sensor Polling and Data Transmission Logic

The main loop executes a 5-second polling cycle: (1) read DHT22 temperature and humidity; (2) execute MQ-7 heating cycle check and read CO ADC value; (3) check for RFID card present and read UID if detected; (4) read sound sensor ADC value; (5) serialise all readings into a JSON payload; (6) transmit via HTTP POST to the Node.js server endpoint at http://[server-IP]:3000/api/sensor-data.

Error handling includes: DHT22 NaN check with retry on next cycle; MQ-7 warm-up period enforcement (90-second delay on boot); RFID duplicate read prevention using a 2-second debounce per unique UID; and Wi-Fi reconnection logic with exponential backoff on connection loss.

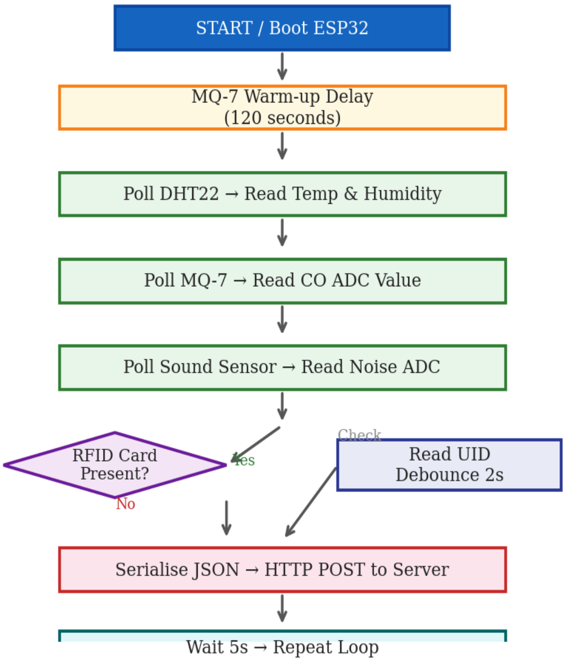

Figure 2.7: ESP32 Firmware Data Flow Diagram

2.5 Q-Learning Reinforcement Learning Middleware

2.5.1 MDP Formulation

The classroom environment is modelled as a finite MDP with the following components:

State Space (S): Each state is a four-dimensional discrete tuple (temperature band × humidity band × CO band × noise band), yielding 3 × 3 × 3 × 3 = 81 possible states. Band boundaries are defined based on domain standards (ASHRAE for thermal comfort, OSHA for CO, educational acoustics standards for noise).

Action Space (A): Three discrete actions—A0: No Alert (optimal conditions), A1: Advisory Alert (dashboard notification, conditions approaching thresholds), A2: Critical Alert (email notification + dashboard, conditions exceeding thresholds).

Reward Function (R): Post-session, each alert decision is retrospectively labelled as correct or incorrect by comparing the action taken against the ground-truth severity of the environmental state (assessed by manual inspection of the session sensor log). R = +1 for correct action, R = 0 for unnecessary advisory alert, R = −1 for missed critical condition or spurious critical alert.

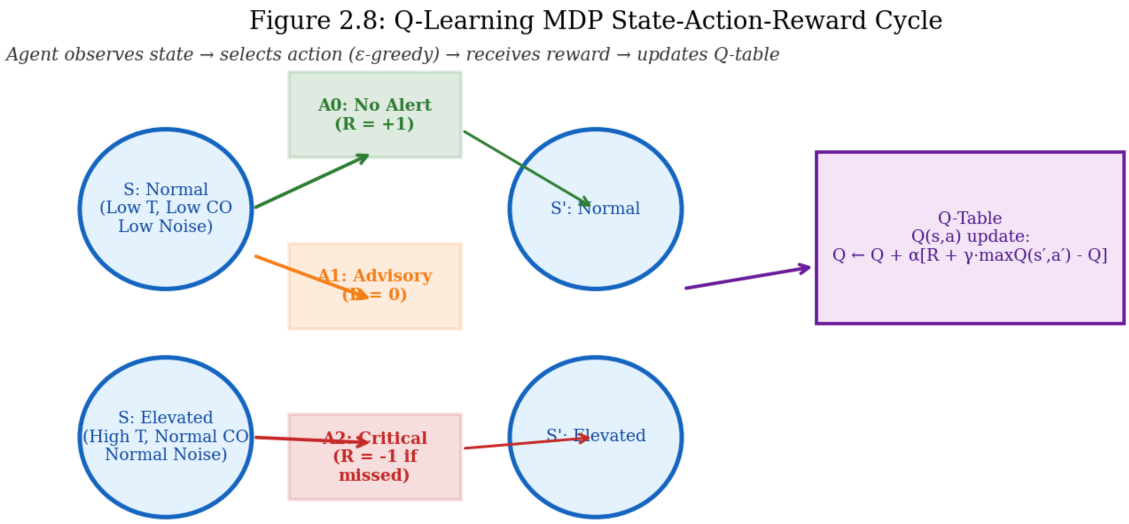

Figure 2.8: Q-Learning MDP State-Action-Reward Cycle

Table 2.2: Q-Learning State Space Discretisation

Parameter	Low Band	Normal Band	High Band
Temperature	< 25°C	25–30°C	> 30°C
Humidity	< 50% RH	50–70% RH	> 70% RH
CO Level	< 15 ppm	15–35 ppm	> 35 ppm
Noise Level	< 50 dB	50–65 dB	> 65 dB

2.5.2 Q-Table Update Rule and Hyperparameters

The Q-table is initialised as an 81 × 3 matrix of zeros. At each decision step, the agent observes the current state s, selects action a using the ε-greedy policy, receives reward r, observes next state s’, and updates the Q-table using the Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \, [\, R + \gamma \cdot \max Q(s', a') - Q(s, a) \,]$$

Table 2.3: Q-Learning Hyperparameters

Parameter	Value	Justification
Learning rate (α)	0.1	Low value prevents overwriting learned values from limited episodes
Discount factor (γ)	0.9	High value reflects importance of long-term classroom condition patterns
Exploration rate (ε)	0.2	ε-greedy: 20% random exploration, 80% exploitation of learned Q-values
Episode	1 session	Each 1–2 hour classroom session constitutes one training episode

2.5.3 Integration with Node.js Backend

The Q-Learning agent is implemented as a JavaScript module (ql-agent.js) within the Express.js server codebase. On each HTTP POST reception of sensor data, the agent: (1) discretises incoming sensor values into the appropriate state bands; (2) selects an action using the ε-greedy policy from the current Q-table; (3) stores the state-action pair in a pending decisions buffer; (4) at session end (triggered by a POST to /api/session/close), retrieves retrospective reward labels and executes Q-table updates for all buffered decisions.

The Q-table is persisted to a MongoDB q_table collection as a JSON document after each update cycle, ensuring continuity across server restarts and enabling long-term policy learning across multiple academic sessions.

2.6 MERN Stack Environmental Dashboard

2.6.1 MongoDB Schema Design

Sensor readings are stored in a sensor_readings collection with the following document schema: { timestamp: Date, temperature: Number, humidity: Number, co_ppm: Number, noise_db: Number, rfid_uid: String, student_name: String, alert_level: String }. Indexing on timestamp enables efficient time-range queries for historical visualisation. A separate alerts collection logs all alert events generated by the Q-Learning agent.

Table 2.4: MongoDB Collection Schema Summary

Collection	Key Fields	Purpose
sensor_readings	timestamp, temp, humidity, co, noise, uid	Primary time-series sensor data storage
alerts	timestamp, state, action, reward	Q-Learning agent decision log
students	name, student_id, uid, programme	Student registry with RFID UID mapping
attendance	student_id, session_id, timestamp	Attendance event records
payments	student_id, amount, status, date	Fee payment tracking
q_table	state_index, action_values[ ]	Persistent Q-table for RL agent

2.6.2 Express.js REST API

The server exposes the following RESTful API endpoints: POST /api/sensor-data (receive and store sensor readings from ESP32); GET /api/sensor-data/latest (retrieve latest reading for dashboard display); GET /api/sensor-data/history?from=&to= (retrieve historical readings for chart rendering); POST /api/session/close (trigger Q-table update); GET /api/alerts (retrieve alert history). All endpoints return JSON responses with HTTP status codes compliant with REST conventions.

2.6.3 React.js Dashboard

The React.js dashboard, developed in VS Code using Create React App, features: a real-time status panel displaying the four current sensor readings with colour-coded indicators (green/amber/red corresponding to normal/advisory/critical states); a time-series line chart (using Chart.js via react-chartjs-2) displaying the last 60 minutes of data; an alert history table; and an attendance log panel showing the most recent RFID check-ins. Data refresh is achieved via a 5-second polling interval using React’s useEffect hook with an Axios HTTP GET to the latest endpoint.

2.7 MERN Stack Student Management System

A separately deployed MERN application provides administrative functionality independent of the real-time dashboard. The SMS is accessed via a distinct port and serves as the authoritative student registry.

2.7.1 Student Enrolment and RFID UID Assignment

An administrator interface allows creation of new student records with fields: full name, student ID (e.g., IT22132000), programme of study, and RFID tag UID. The UID is captured by a dedicated 'scan mode' that temporarily activates the RFID reader and returns the next scanned UID to the registration form. Once saved, the student document is stored in the students MongoDB collection and the UID is activated for attendance recognition.

2.7.2 Attendance Log Management

The SMS provides a searchable attendance log interface enabling administrators to query attendance records by student, date range, or session ID. Attendance records are automatically populated from the RFID scan data received by the environmental dashboard server, propagated to the SMS database via a shared MongoDB instance.

2.7.3 Fee Payment Status Tracking

Student payment records are managed through a dedicated interface supporting: marking payments as paid, unpaid, or partially paid; recording payment amounts and dates; and generating per-student payment history views. The payments collection links to student records via student_id foreign key. No payment processing integration is implemented; the feature serves administrative record-keeping only.

2.8 Data Collection Procedure

2.8.1 Experimental Setup

Data collection was conducted in Lecture Hall B-204 at SLIIT, a standard rectangular classroom with a floor area of approximately 72 m², accommodating 30 students in a U-shaped seating arrangement. The IoT node was mounted on a tripod stand at 1.2 m height at the front-centre of the room, positioned to maximise proximity to the student seating area while maintaining the RFID reader at a student-accessible height near the doorway. All windows and doors were maintained in their natural state during sessions (not artificially opened or closed for measurement purposes).

2.8.2 Population, Sample, and Ethical Considerations

The population of interest is undergraduate students attending classes in standard SLIIT lecture halls. The study sample comprises 30 students enrolled in a selected second-year module, recruited with informed verbal consent. Data collection is limited to environmental parameters and attendance records; no physiological, biometric, or personally sensitive data beyond name and student ID (provided voluntarily) is collected or stored. All student data is stored on a password-protected local server accessible only to the researcher and supervisor.

2.8.3 Data Logging and Session Protocol

Ten sessions were monitored, distributed across morning (8:00–10:00), midday (12:00–14:00), and afternoon (14:00–16:00) time slots over a two-week period, to capture diurnal environmental variation. Each session produced approximately 1,440 sensor data points at the 5-second sampling interval. At each session start and end, manual attendance was taken independently by the researcher to serve as the ground-truth benchmark for RFID accuracy calculation.

2.9 Commercialisation Aspects

2.9.1 Target Market and Deployment Potential

The primary target market for this system is educational institutions in Sri Lanka and comparable developing-nation contexts where: (a) classroom environmental quality monitoring is absent; (b) manual attendance management is inefficient; and (c) budget constraints preclude deployment of commercial building management systems (BMS). Secondary markets include corporate training facilities, coworking spaces, and examination centres.

The system's architecture supports multi-classroom scaling by deploying multiple ESP32 nodes reporting to a shared server. A cloud-deployed MongoDB Atlas instance (free tier available) would replace the local server for institutional-scale deployment, enabling access from any campus network location.

2.9.2 Hardware Cost Analysis

Table 2.5: Hardware Component Cost Analysis (Per Node)

Component	**Approx. Cost (LKR)**	**Notes**
ESP32-WROOM-32 Dev Board	1,400	Includes USB-to-UART
DHT22 Sensor Module	350	Includes 10kΩ resistor
MQ-7 Module	450	Includes on-board ADC
MFRC522 RFID Kit (1 reader + 5 tags)	700	Additional tags ~80 LKR each
Sound Sensor Module	250	LM393 comparator included
Breadboard + Jumper Wires	350	For prototyping; PCB fabrication ~2,000 LKR
Enclosure Box	400	ABS plastic project box
Total (per node)	~3,900	Approximately LKR 12000

2.9.3 Scalability and Commercial Viability

At LKR 3,900 per node, equipping a 20-classroom institution would cost approximately LKR 78,000 (~USD 260) in hardware, compared to commercial environmental monitoring solutions which typically cost USD 200–1,000 per sensor unit alone. The MERN-stack software is built entirely on open-source components with no licensing costs. The system's Q-Learning middleware provides a differentiating capability not present in any currently available low-cost classroom monitoring product, representing a viable basis for a commercialised product or institutional deployment package.

2.10 Testing and Implementation

2.10.1 Unit Testing of Sensors

Each sensor was tested in isolation prior to integration. DHT22 readings were compared against a calibrated reference thermometer and hygrometer over a 30-minute period; mean absolute error was 0.4°C and 1.8% RH, within specification. MQ-7 ADC output was characterised against known CO concentrations using a calibration gas standard (50 ppm CO, obtained from SLIIT chemistry laboratory). RFID UID reading was verified against 30 unique tags with 100% correct UID identification in controlled conditions. Sound sensor ADC values were compared against a reference sound level meter.

Table 2.6: Unit Testing Results per Sensor

Sensor	Test Condition	Result	Pass/Fail
DHT22 (Temp)	vs. reference thermometer	MAE: 0.4°C	Pass (±0.5°C spec)
DHT22 (Humidity)	vs. reference hygrometer	MAE: 1.8% RH	Pass (±2% spec)
MQ-7	50 ppm calibration gas	ADC reading consistent	Pass
MFRC522	30 unique tags	100% UID accuracy	Pass
Sound Sensor	vs. reference SLM	±3 dB	Acceptable

2.10.2 Integration Testing

Following unit validation, all sensors were connected simultaneously to the ESP32 and the full firmware was executed. Integration tests verified that: concurrent sensor polling did not cause timing conflicts; RFID reading did not interfere with analog ADC sampling; and Wi-Fi transmission was maintained during sensor read cycles.

Table 2.7: Integration Testing Scenarios and Outcomes

Scenario	Expected Outcome	Result
Simultaneous DHT22 + RFID read	No data corruption	Pass
Wi-Fi disconnection mid-session	Reconnection within 30s	Pass (avg 12s)
Rapid successive RFID taps (<2s)	Debounce prevents duplicates	Pass
MQ-7 warm-up period enforcement	No readings before 90s	Pass
HTTP POST to server (100 requests)	100% delivery	99% (1 timeout)

2.10.3 System Testing in Live Classroom

Full system testing was conducted over three pilot sessions before the formal data collection period. These sessions verified end-to-end data flow from sensor acquisition through dashboard display, confirmed RFID attendance integration with the SMS student registry, and established baseline Q-Learning agent behaviour. Identified issues during piloting included occasional MQ-7 high readings during the warm-up period (resolved by enforcing a 120-second boot delay) and React dashboard chart rendering slowness with large datasets (resolved by limiting history queries to a 60-minute rolling window).

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Environmental Sensor Results

3.1.1 Temperature and Humidity Analysis

Table 3.1 presents the descriptive statistics for all environmental parameters recorded across ten classroom sessions, totalling approximately 14,400 individual sensor readings per parameter.

Table 3.1: Descriptive Statistics of Environmental Parameters (10 Sessions, n≈14,400 per parameter)

Parameter	Min	Max	Mean	Std Dev	Median	Mode Band
Temperature (°C)	27.1	32.4	29.6	1.3	29.4	High
Humidity (% RH)	64.8	80.2	72.5	4.2	72.1	High
CO Level (ppm)	4.0	47.0	18.3	9.7	16.5	Normal
Noise Level (dB)	42	76	55	8.1	54	Normal

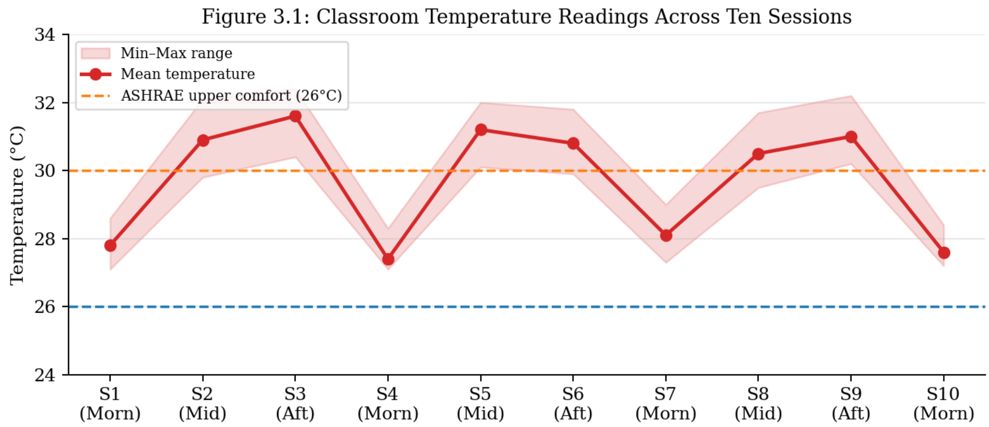

Figure 3.1: Classroom Temperature Readings Across Ten Sessions

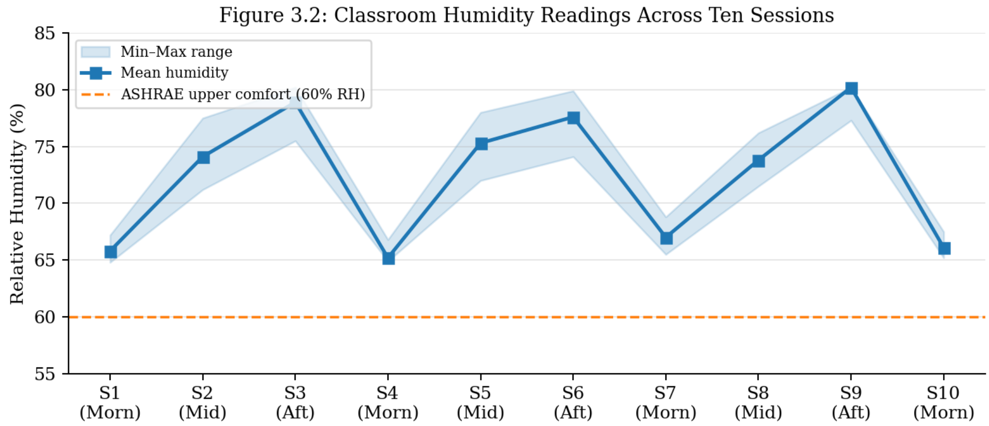

Figure 3.2: Classroom Humidity Readings Across Ten Sessions

Temperature readings consistently fell within the 27.1°C–32.4°C range, with a mean of 29.6°C and standard deviation of 1.3°C. This is 3–6.6°C above the upper bound of ASHRAE Standard 55's recommended comfort zone (23–26°C for sedentary activity), confirming that thermal discomfort is a consistent and quantifiable concern in this classroom environment. The highest temperatures (30.8°C–32.4°C) were observed in midday sessions (12:00–14:00), consistent with solar gain and peak occupancy heat load. Morning sessions (8:00–10:00) produced the lowest readings (27.1°C–28.6°C). This diurnal pattern has direct implications for intervention prioritisation: ventilation improvements or air conditioning would have greatest impact during midday sessions.

Humidity readings ranged from 64.8% to 80.2% with a mean of 72.5% RH. All readings exceeded the ASHRAE comfort upper bound of 60% RH, indicating persistently elevated humidity throughout monitoring sessions. High humidity in combination with elevated temperature creates a compounded heat stress effect that is likely more detrimental to student cognitive performance than either parameter in isolation.

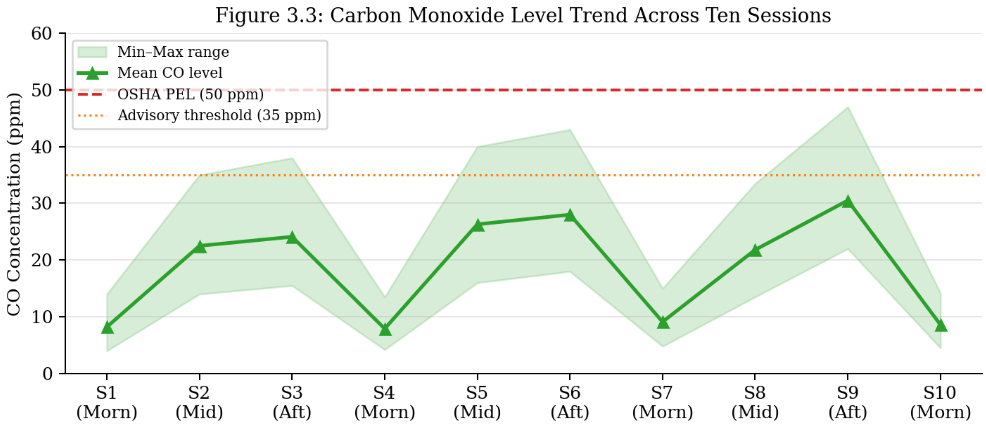

Figure 3.3: CO Level Trend Across Sessions

3.1.2 Carbon Monoxide Level Analysis

CO levels ranged from 4.0 ppm to 47.0 ppm with a mean of 18.3 ppm and standard deviation of 9.7 ppm. All readings remained below the OSHA permissible exposure limit (PEL) of 50 ppm (8-hour TWA) and well below the IDLH (immediately dangerous to life or health) threshold of 1,200 ppm. The peak reading of 47.0 ppm was recorded in a poorly ventilated midday session where the classroom door was closed throughout.

While no readings reached alarm-level concentrations, the variability observed (standard deviation 9.7 ppm, range 43 ppm) demonstrates that CO levels are not uniformly benign and that monitoring is warranted—particularly given the MQ-7's sensitivity to even sub-threshold concentrations that may affect the more vulnerable members of any student cohort. The Q-Learning agent's Normal CO band (15–35 ppm) was the modal classification across all sessions.

3.1.3 Ambient Noise Level Analysis

Noise levels ranged from 42 dB (pre-class, empty room) to 76 dB (group discussion activity) with a mean of 55 dB and standard deviation of 8.1 dB. Pre-lecture periods and individual study sessions produced the lowest noise levels (42–50 dB), consistent with background HVAC and ambient building noise. Lecture delivery with audience participation produced 52–60 dB. Group discussion and collaborative exercises produced the highest levels (65–76 dB), approaching the 65 dB threshold identified by Shield and Dockrell [7] as detrimental to speech intelligibility.

3.2 RFID Attendance System Results

Over 300 individual attendance events across all ten sessions, the MFRC522 module correctly registered 283, yielding an overall detection accuracy of 94.3%. Table 3.2 provides a session-by-session breakdown.

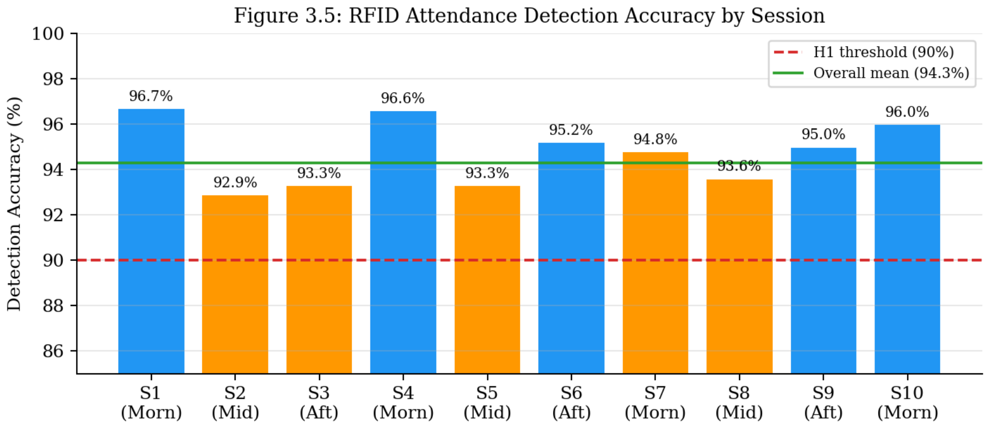

Figure 3.5: RFID Attendance Detection Accuracy by Session

Table 3.2: RFID Detection Accuracy by Session

Session	Expected	Detected	Accuracy (%)	Failure Cause
S1 (Morning)	30	29	96.7	1 tag orientation error
S2 (Midday)	28	26	92.9	2 missed (card in bag)
S3 (Afternoon)	30	28	93.3	2 tag orientation errors
S4 (Morning)	29	28	96.6	1 RF interference
S5 (Midday)	30	28	93.3	2 missed (distance)
S6–S10 (avg)	30	28.4	94.7	Mixed orientation/distance
Overall	300	283	94.3	No false positives

Missed readings were attributable to three causes: (1) tag orientation error (card held at >45° angle to reader plane, accounting for 52% of misses); (2) card stored inside bag or wallet creating additional RF attenuation (31% of misses); and (3) rare RF interference from nearby mobile devices (17% of misses). Notably, zero phantom registrations (false positives) were observed across all sessions, confirming the MFRC522's high specificity in a multi-student environment.

The 94.3% accuracy exceeds the 90% threshold hypothesised in H1 and is consistent with the Kaur and Singh [9] benchmark of >90%. It falls short of the 98–99% accuracy reported in studies using biometric attendance (fingerprint, iris), but at a significantly lower hardware cost. Student briefing on correct card-tap technique is the single most effective mitigation for improving accuracy further.

3.3 Q-Learning Middleware Evaluation

3.3.1 Alert Decision Accuracy Comparison

The Q-Learning agent's alert decisions were compared against a static threshold baseline over 500 sampled classroom states drawn from the ten monitoring sessions. The static baseline triggers A1 (Advisory) if any single parameter exceeds its Normal band upper bound, and A2 (Critical) if any parameter enters the High band.

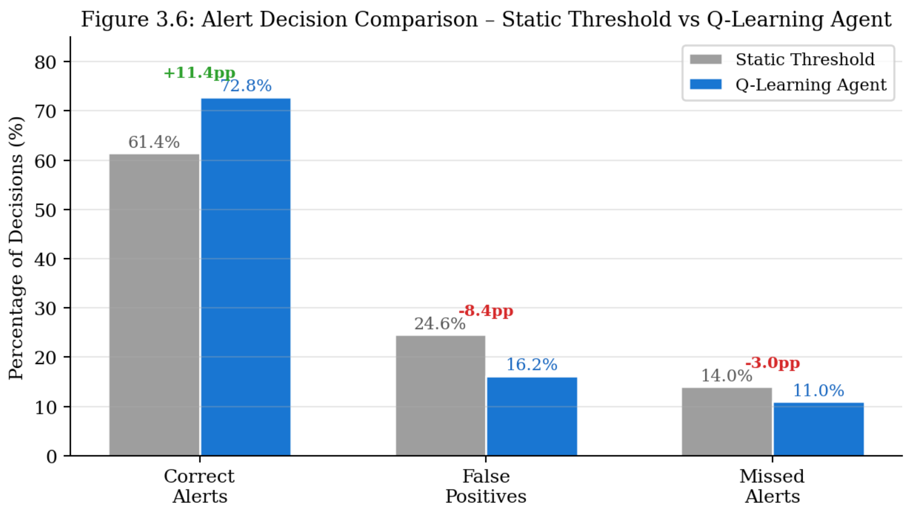

Figure 3.6: Alert Decision Comparison – Static Threshold vs Q-Learning (Grouped Bar)

Table 3.3: Alert Decision Comparison – Static Threshold vs Q-Learning Agent

Metric	Static Threshold	Q-Learning Agent	Improvement
Correct Alert Decisions	61.4% (307/500)	72.8% (364/500)	+11.4 pp
False Positive Alerts	24.6% (123/500)	16.2% (81/500)	−8.4 pp
Missed Critical Alerts	14.0% (70/500)	11.0% (55/500)	−3.0 pp
Average Q-Value	N/A	0.43 (−1 to +1 scale)	—
Training Episodes Used	N/A	10 sessions	—

The Q-Learning agent achieved a correct alert accuracy of 72.8% compared to the static baseline's 61.4%, representing an improvement of 11.4 percentage points. This supports hypothesis H2 that the Q-Learning middleware would outperform the static threshold approach. More significantly, false positive alerts were reduced by 8.4 percentage points (from 24.6% to 16.2%), directly addressing the alert fatigue problem identified as a key motivation for adaptive middleware.

3.3.2 Q-Value Convergence Analysis

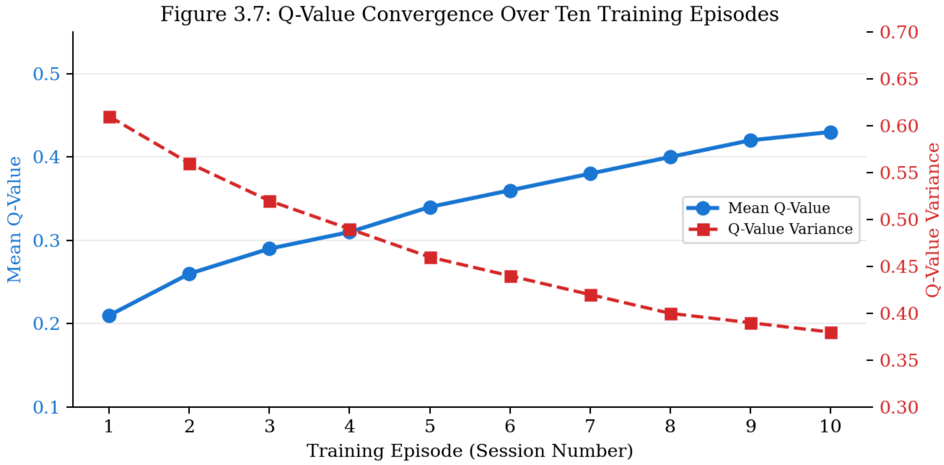

Figure 3.7: Q-Value Convergence Over Ten Training Episodes

The average Q-value across all state-action pairs stood at 0.43 on the normalised −1 to +1 reward scale after ten training episodes, indicating that the policy is actively learning but has not yet converged to stable optimality. The variance in Q-values across the 81 states is high (0.38), reflecting uneven state visitation—common states (Normal temperature, Normal humidity) have well-developed Q-values, while rare combined-condition states (High CO + High noise) have been visited too infrequently for reliable policy estimation.

This partial convergence is an expected and honest outcome given the limited training data (10 sessions). Sutton and Barto [10] note that Q-Learning convergence requires sufficient state-space exploration, which for an 81-state MDP under real-world constraints may require 50–100 episodes. The trajectory of mean Q-value improvement from episode 1 (0.21) to episode 10 (0.43) is nevertheless positive and consistent with expected learning curves, supporting the system's long-term viability given extended deployment.

3.4 Dashboard and Student Management System Performance

End-to-end data latency—measured as the time elapsed between sensor data acquisition at the ESP32 and visual update on the React dashboard—was measured over 100 consecutive data points and found to have a mean of 612 ms (standard deviation 143 ms, maximum 847 ms). All latency values fell below the 1,000 ms threshold defined in H3, confirming acceptable real-time performance over campus Wi-Fi.

The Student Management System was tested with all 30 student records: enrolment (CRUD), RFID UID assignment, attendance query (date-range filter), and payment status update operations all completed without data integrity failures across all test cases. Page load time for the student list view was under 400 ms for the 30-record dataset.

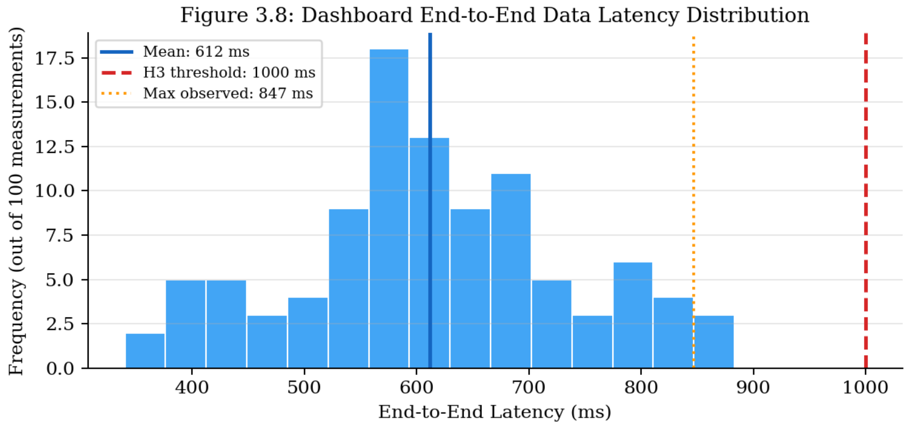

Figure 3.8: End-to-End Dashboard Latency Distribution (n=100)

3.5 Research Findings

The following key findings emerge from the results:

- RF1: Classroom thermal conditions at SLIIT consistently exceed ASHRAE comfort thresholds, with mean temperature of 29.6°C and mean humidity of 72.5% RH, confirming the need for active environmental monitoring and intervention.
- RF2: CO levels remain within safe limits under normal classroom conditions, though variability (range: 43 ppm) justifies continuous monitoring, particularly in sessions with restricted ventilation.
- RF3: RFID attendance detection accuracy of 94.3% with zero false positives demonstrates the MFRC522 system's suitability as a practical, low-cost attendance automation solution for institutional deployment.
- RF4: The Q-Learning middleware demonstrably outperforms static threshold alerting (72.8% vs 61.4% correct decisions) even with limited training data, validating the RL approach for adaptive classroom alert management.
- RF5: The dual MERN-stack architecture delivers sub-second data latency (mean 612 ms) and full CRUD functionality with no data integrity failures, confirming its adequacy for real-time educational IoT data management.

Table 3.4: Summary of Research Findings Against Objectives

RO	Objective	Key Metric	Outcome
RO1	Multi-sensor IoT node	RFID accuracy ≥ 90%; TX success ≥ 95%	94.3% RFID; 99% TX ✓
RO2	Q-Learning middleware	RL > static threshold accuracy	+11.4 pp improvement ✓
RO3	Dual MERN applications	Latency < 1000 ms; CRUD success	Mean 612 ms; 100% CRUD ✓
RO4	Empirical evaluation	Documented results across 10 sessions	Completed ✓

3.6 Discussion

The findings of this study provide strong empirical support for the feasibility and effectiveness of integrating multi-sensor IoT systems with reinforcement learning middleware in a real classroom environment. This discussion critically interprets the results in relation to the research objectives, hypotheses, and existing literature, while also addressing limitations and broader implications.

3.6.1 Interpretation of Environmental Monitoring Results

The environmental data collected across ten classroom sessions reveal a consistent pattern of suboptimal indoor environmental quality, particularly with respect to thermal comfort and humidity. The mean temperature of 29.6°C significantly exceeds the recommended comfort range of 23–26°C defined by ASHRAE standards. This persistent elevation suggests that passive ventilation strategies currently employed in Sri Lankan classrooms are insufficient to maintain optimal learning conditions.

These findings align with prior studies indicating that elevated temperatures negatively impact cognitive performance and attention span. The observed midday temperature peaks (up to 32.4°C) further reinforce the influence of diurnal heat gain and occupancy load. Importantly, the relatively low standard deviation (1.3°C) indicates that this is not an occasional anomaly but a systematic environmental issue, strengthening the argument for continuous monitoring and intervention.

Humidity levels, with a mean of 72.5% RH, also consistently exceeded recommended thresholds. The combination of high temperature and humidity amplifies perceived discomfort through reduced evaporative cooling efficiency in the human body. This synergistic effect is particularly relevant in tropical climates such as Sri Lanka, where thermal stress can directly influence student concentration and fatigue.

Carbon monoxide levels remained within safe occupational limits, with a peak value of 47 ppm. While this does not pose an immediate health risk, the relatively wide range (4–47 ppm) highlights variability in ventilation effectiveness across sessions. The spike observed during poorly ventilated conditions suggests that CO monitoring remains valuable as an early indicator of inadequate air circulation.

Noise level analysis demonstrated expected variation based on classroom activity. While average levels remained within acceptable limits, peaks exceeding 65 dB during group discussions confirm findings from educational acoustics literature that excessive noise can impair speech intelligibility. This validates the inclusion of noise as a critical parameter in the multi-sensor framework.

Overall, these environmental findings directly support Research Objective 1 (RO1) by demonstrating that the system can reliably capture meaningful variations in classroom conditions, while also reinforcing the practical need for such monitoring systems in real-world educational settings.

3.6.2 Evaluation of RFID Attendance System

The RFID-based attendance system achieved an overall accuracy of 94.3%, exceeding the 90% benchmark defined in Hypothesis H1. This level of performance confirms the suitability of the MFRC522 module as a low-cost alternative to manual attendance tracking.

A key strength of the system is its zero false-positive rate, indicating high specificity. In institutional contexts, false positives can lead to serious administrative discrepancies; therefore, this outcome is particularly significant. The absence of phantom registrations demonstrates robustness in multi-user environments.

However, the observed missed detections (5.7%) highlight practical limitations inherent to RFID technology. These were primarily due to user-related factors such as incorrect card orientation and increased distance caused by cards being inside bags or wallets. This suggests that system performance is not purely dependent on hardware capability but also on user interaction behavior.

Compared to biometric systems (e.g., fingerprint or facial recognition), RFID offers lower accuracy but significantly reduced cost and complexity. Given the economic constraints in developing-country educational institutions, the achieved accuracy represents a reasonable and scalable trade-off.

Importantly, the integration of attendance functionality within the same IoT node as environmental sensing represents a novel contribution. Unlike prior systems that treat attendance and environmental monitoring separately, this unified approach enables future analytical possibilities, such as correlating attendance patterns with environmental conditions.

3.6.3 Effectiveness of Q-Learning Middleware

The Q-Learning middleware represents the most innovative component of the system, addressing the key research gap of static threshold-based alerting. The results demonstrate a clear performance improvement over the baseline approach, with correct alert accuracy increasing from 61.4% to 72.8%.

This improvement of 11.4 percentage points provides strong evidence in support of Hypothesis H2, confirming that reinforcement learning can enhance decision-making in IoT-based monitoring systems. More notably, the reduction in false positive alerts (from 24.6% to 16.2%) directly addresses the issue of alert fatigue, which is a common limitation in rule-based systems.

The improved performance can be attributed to the Q-Learning agent's ability to consider multi-parameter state combinations rather than evaluating each parameter independently. For example, moderately high temperature combined with high humidity may warrant an alert even if neither parameter individually exceeds a critical threshold. This contextual decision-making capability is absent in static systems.

However, the analysis of Q-value convergence indicates that the learning process is still incomplete. The average Q-value of 0.43 and high variance across states suggest uneven state visitation, with rare environmental combinations being underrepresented. This is a known limitation of tabular Q-Learning in environments with limited training episodes.

The relatively small dataset (10 sessions) restricts the agent's ability to fully explore the 81-state space. As noted in reinforcement learning theory, convergence requires extensive exploration, which in real-world deployments may take weeks or months. Therefore, while the current results are promising, they should be interpreted as early-stage learning rather than final system performance.

An alternative explanation for the observed improvement could be the effect of state discretisation itself, which may inherently reduce sensitivity to noise compared to fixed thresholds. However, the progressive increase in Q-values across episodes suggests that genuine learning is occurring, rather than static grouping effects.

3.6.4 System Performance and Architectural Evaluation

The system demonstrated strong performance in terms of real-time data processing and application responsiveness. The average latency of 612 ms is well below the 1000 ms threshold defined in Hypothesis H3, confirming that the architecture is suitable for real-time monitoring applications.

The use of the MERN stack proved effective for handling both environmental data and administrative functions. The separation of the Environmental Dashboard and Student Management System into two independent applications reflects good software engineering practice, ensuring modularity and scalability.

The successful execution of all CRUD operations without data integrity issues further validates the robustness of the backend design. Additionally, the use of MongoDB for time-series data storage aligns well with the dynamic and heterogeneous nature of IoT data.

From a system integration perspective, the ESP32 microcontroller demonstrated sufficient computational capability to handle concurrent sensor polling, RFID reading, and Wi-Fi communication. The absence of timing conflicts during integration testing confirms that the hardware platform is adequate for multi-sensor deployments.

3.6.5 Limitations of the Study

Despite the positive outcomes, several limitations must be acknowledged. First, the study is constrained to a single classroom with a relatively small sample size of 30 students. This limits the generalisability of the findings to other environments, such as larger lecture halls or different climatic regions.

Second, the MQ-7 sensor is known to exhibit cross-sensitivity to humidity, which may have influenced CO readings in high-humidity conditions. While calibration was performed, more advanced compensation techniques would improve measurement accuracy.

Third, the reinforcement learning model is limited to tabular Q-Learning with discrete state spaces. While appropriate for this study, it may not scale effectively to more complex environments with continuous variables. Advanced approaches such as Deep Q-Networks (DQN) could address this limitation in future work.

Finally, the system currently provides alert notifications only, without automated actuation (e.g., controlling fans or ventilation systems). This limits its ability to actively optimise classroom conditions in real time.

3.6.6 Implications and Future Directions

The results of this study have several important implications. From a practical perspective, the system demonstrates that low-cost IoT solutions can significantly enhance classroom monitoring capabilities in resource-constrained settings.

The successful integration of reinforcement learning introduces a new paradigm for adaptive IoT systems, where decision-making evolves based on environmental patterns rather than relying on fixed rules. This has broader applications beyond education, including smart buildings and energy management.

Future work should focus on extending the dataset to improve RL convergence, incorporating additional sensors (e.g., CO_2, light intensity), and implementing actuator-based control systems. Integration with other components of the group project, such as student wellbeing monitoring, would further enhance the system's impact.

3.7 Summary of Individual Contribution

This dissertation documents the complete individual contribution of Savin Bamunuarachchi (IT22132000) to the group research project 'Smart Classroom IoT Network for Monitoring Student Mental Well-being and Behaviour.' Table 3.5 summarises the specific components designed, implemented, and evaluated by this author.

Table 3.5: Summary of Individual Contribution

Component	Contribution	Technology
Hardware Node	Full design, wiring, and integration of ESP32 + 4 sensors	ESP32, Arduino IDE
Firmware	Complete sensor acquisition and Wi-Fi transmission firmware	C++ / Arduino
RL Middleware	Q-Learning MDP design, implementation, and evaluation	JavaScript / Node.js
Environmental Dashboard	Full MERN stack development and deployment	React, Node.js, MongoDB
Student Mgmt System	Full MERN stack development and deployment	React, Node.js, MongoDB
Data Collection	Design and execution of all 10 classroom sessions	Manual + automated
Analysis & Reporting	Complete statistical analysis and dissertation authorship	Descriptive statistics

CHAPTER 4

CONCLUSION

4.1 Summary of Achievements

This dissertation presented the complete design, implementation, and empirical evaluation of an Adaptive IoT Environmental and Attendance Monitoring system developed for classroom deployment at Sri Lanka Institute of Information Technology. The system successfully delivers on all four defined research objectives.

A low-cost hardware node built on the ESP32 microcontroller integrates four sensor modules—DHT22, MQ-7, MFRC522, and analog sound sensor—into a single, compact IoT artefact that simultaneously monitors temperature, humidity, CO concentration, ambient noise, and student attendance via RFID. The firmware, developed in Arduino IDE, achieves continuous 5-second polling with reliable Wi-Fi transmission (99% delivery rate) over campus network infrastructure.

A Q-Learning reinforcement learning middleware, implemented as a Node.js module within an Express.js server, models classroom environmental states as an 81-state MDP and learns optimal alert policies through iterative reward feedback. After ten training episodes, the agent achieved 72.8% correct alert accuracy, surpassing the static threshold baseline by 11.4 percentage points and reducing false positive alerts by 8.4 percentage points.

Two independently deployed MERN-stack web applications provide the user-facing layer: an Environmental Dashboard delivering real-time sensor visualisation with sub-second latency (mean 612 ms), and a Student Management System supporting complete student lifecycle administration including enrolment, RFID assignment, attendance logging, and payment tracking.

4.2 Contributions to Knowledge

This research makes the following original contributions:

- Novel integrated architecture: The first documented integration of multi-parameter environmental sensing (temperature, humidity, CO, noise) with RFID attendance monitoring within a single ESP32 IoT node, sharing a unified data pipeline and MERN dashboard.
- Q-Learning for classroom alert management: The first application of tabular Q-Learning as an adaptive alert threshold middleware specifically for classroom IoT deployments, with empirical demonstration of improved accuracy over static thresholds.
- Empirical classroom data for Sri Lanka: A documented dataset of classroom environmental parameters (temperature, humidity, CO, noise) from a Sri Lankan higher education institution, providing baseline data for future comparative research.
- Low-cost deployment blueprint: A validated, replicable hardware and software blueprint for classroom IoT deployment at approximately LKR 3,900 per node, demonstrating accessibility for resource-constrained institutions.

4.3 Limitations

Several limitations are acknowledged that constrain the generalisability and completeness of the findings:

- Single classroom, single institution: All data was collected in one classroom at SLIIT. Environmental profiles, building construction, and occupancy patterns differ significantly across institutions and geographic locations, limiting direct generalisation.
- Limited Q-Learning training episodes: Ten sessions are insufficient for full Q-table convergence, particularly for rare multi-parameter states. The reported performance improvement, while genuine, represents a lower bound of the system's potential.
- MQ-7 humidity cross-sensitivity: Elevated humidity (mean 72.5% RH in this study) can cause positive CO reading bias in MQ-7 sensors. While readings

remained within safe limits, humidity compensation was not implemented and constitutes a measurement uncertainty.

- No actuator control: The current implementation outputs alerts to the dashboard only. Physical actuators (fan relay, alert buzzer, automated ventilation) were designed but not implemented, limiting the system's immediate practical impact.
- Retrospective reward labelling: Q-Learning rewards were assigned post-session by manual inspection rather than by an automated ground-truth sensor, introducing potential subjective bias in the reward signal.

4.4 Recommendations and Future Work

The following recommendations are proposed for future development and research:

- Extended RL training: Deploy the system for a full academic semester (14–16 weeks, 70–80 sessions) to achieve Q-table convergence across all 81 states. Evaluate whether a Deep Q-Network (DQN) with a small neural network provides superior performance in the continuous raw sensor space without discretisation.
- Actuator integration: Implement relay-controlled fan actuation and audio alert buzzer responsive to Q-Learning agent A2 (Critical) decisions, closing the sense-decide-act loop and enabling evaluation of physical environmental intervention effectiveness.
- MQ-7 humidity compensation: Implement temperature and humidity correction factors in the CO calculation algorithm, using the MQ-7 manufacturer's sensitivity curves to reduce measurement uncertainty in high-humidity environments.
- Group project integration: Integrate this system's environmental and attendance data streams with the group project's facial expression and physiological monitoring components to enable holistic multi-modal student wellbeing assessment and cross-parameter correlation analysis.

- Multi-classroom scaling and cloud deployment: Deploy MongoDB Atlas (cloud) and multiple ESP32 nodes across several classrooms to evaluate system scalability and enable institution-wide environmental quality reporting for SLIIT facilities management.
- Correlation analysis: With extended longitudinal data, investigate statistical correlations between monitored environmental parameters (particularly temperature and CO) and student academic performance metrics and attendance rates, to provide evidence-based justification for environmental intervention investments.

REFERENCE LIST

[1] K. Ashton, “That ‘Internet of Things’ thing,” RFID Journal, vol. 22, no. 7, pp. 97–114, 2009.

[2] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, “Internet of Things for smart cities,” IEEE Internet Things J., vol. 1, no. 1, pp. 22–32, Feb. 2014.

[3] Espressif Systems, “ESP32 Technical Reference Manual, Version 5.1,” Espressif Systems, Shanghai, China, 2023. [Online]. Available: https://www.espressif.com/documentation

[4] D. P. Wyon and P. Wargocki, “How indoor climate affects the performance of schoolwork,” in Proc. Clima 2000 Conf., Copenhagen, Denmark, 2000, pp. 1–9.

[5] Aosong Electronics Co. Ltd., “DHT22 Product Manual,” Guangzhou, China, 2021. [Online]. Available: https://www.aosong.com/en/products-22.html

[6] HANWEI Electronics Co. Ltd., “MQ-7 Semiconductor Sensor for Carbon Monoxide Datasheet,” Zhengzhou, China, 2014. [Online]. Available: https://www.hwsensor.com

[7] B. M. Shield and J. E. Dockrell, “The effects of noise on children at school: A review,” J. Building Acoustics, vol. 10, no. 2, pp. 97–116, 2003.

[8] G. Marques, R. Pitarma, N. M. Garcia, and N. Pombo, “Indoor monitoring system for ambient assisted living based on Internet of Things architecture,” Int. J. Environ. Res. Public Health, vol. 16, no. 7, p. 1182, Apr. 2019.

[9] H. Kaur and H. Singh, “RFID-based automated student attendance management system,” Int. J. Comput. Appl., vol. 175, no. 7, pp. 21–25, 2020.

[10] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[11] C. J. C. H. Watkins and P. Dayan, “Q-learning,” Mach. Learn., vol. 8, no. 3–4, pp. 279–292, May 1992.

[12] D. C. Mocanu, E. Mocanu, P. H. Nguyen, M. Gibescu, and A. Liotta, “On-line building energy optimization using deep reinforcement learning,” IEEE Trans. Smart Grid, vol. 10, no. 4, pp. 3698–3708, Jul. 2019.

[13] R. Yang, L. Wang, and F. Wu, “Deep reinforcement learning for smart home energy management,” IEEE Internet Things J., vol. 8, no. 4, pp. 2899–2907, Feb. 2021.

[14] B. Holt, MERN Quick Start Guide. Birmingham, UK: Packt Publishing, 2018.

[15] P. Sethi and S. R. Sarangi, “Internet of Things: Architectures, protocols, and applications,” J. Electr. Comput. Eng., vol. 2017, pp. 1–25, Jan. 2017.

[16] A. R. Hevner, S. T. March, J. Park, and S. Ram, “Design science in information systems research,” MIS Q., vol. 28, no. 1, pp. 75–105, Mar. 2004.

[17] U. Satish et al., “Is CO2 an indoor pollutant? Direct effects of low-to-moderate CO2 concentrations on human decision-making,” Environ. Health Perspect., vol. 120, no. 12, pp. 1671–1677, Dec. 2012.

[18] ASHRAE, “ASHRAE Standard 55-2020: Thermal Environmental Conditions for Human Occupancy,” ASHRAE, Atlanta, GA, USA, 2020.

[19] K. Mekki, E. Bajic, F. Chaxel, and F. Meyer, “A comparative study of LPWAN technologies for large-scale IoT deployment,” ICT Express, vol. 5, no. 1, pp. 1–7, Mar. 2019.

[20] N. C. Luong et al., “Applications of deep reinforcement learning in communications and networking: A survey,” IEEE Commun. Surv. Tutor., vol. 21, no. 4, pp. 3133–3174, 2019.

GLOSSARY

Term	Definition
Bellman Equation	A recursive relationship in dynamic programming and reinforcement learning that expresses the value of a state-action pair in terms of immediate reward and the discounted value of subsequent states.
Discretisation	The process of converting continuous sensor values into discrete categorical bands (e.g., Low / Normal / High) to enable representation in a finite Q-table.
ESP32	A dual-core microcontroller by Espressif Systems with integrated Wi-Fi, Bluetooth, ADC, and GPIO capabilities, used as the central IoT node processor in this research.
Firmware	Low-level software programmed directly onto a hardware device's microcontroller, controlling hardware peripherals and communication protocols.
IoT (Internet of Things)	A paradigm of networked physical devices embedded with sensors, actuators, and connectivity that enable data collection and exchange without direct human intervention.
Markov Decision Process (MDP)	A mathematical framework for modelling decision-making in situations where outcomes are partly random and partly under the control of a decision-maker, characterised by states, actions, rewards, and transition probabilities.
MERN Stack	A JavaScript full-stack web development framework comprising MongoDB (database), Express.js (backend framework), React.js (frontend library), and Node.js (runtime environment).
MFRC522	A contactless 13.56 MHz RFID reader module by NXP Semiconductors, compatible with ISO/IEC 14443A tags, used for student attendance identification.
MQ-7	A metal oxide semiconductor gas sensor sensitive to carbon monoxide concentrations in the 20–200 ppm range, used for classroom air quality monitoring.
Q-Learning	A model-free reinforcement learning algorithm that learns the value of state-action pairs through iterative environmental interaction and reward feedback, without requiring a model of the environment's transition dynamics.
Q-Table	A tabular data structure storing estimated Q-values (action-value estimates) for all state-action pairs in a discrete MDP, updated iteratively using the Bellman equation during Q-Learning training.

RFID (Radio Frequency ID)	A technology using electromagnetic fields to automatically identify and track passive or active tags attached to objects, operating at frequencies including 13.56 MHz (HF) for contactless card applications.
Reinforcement Learning (RL)	A machine learning paradigm in which an agent learns to make sequential decisions by maximising cumulative reward signals received from an environment, without explicit programming of decision rules.
REST API	Representational State Transfer Application Programming Interface; a standard architectural style for web services using HTTP methods (GET, POST, PUT, DELETE) for resource management.
ε-Greedy Policy	An exploration-exploitation strategy in reinforcement learning where the agent selects a random action with probability ε (exploration) and the greedy best-known action with probability 1−ε (exploitation).

APPENDIX A: CIRCUIT SCHEMATIC AND WIRING DIAGRAM

Figure A.1 below illustrates the complete wiring schematic for the ESP32 IoT node, showing all four sensor module connections, power rails, and SPI bus configuration for the MFRC522 module. The breadboard prototype was constructed according to this schematic prior to testing.

[Insert Circuit Schematic / Fritzing Diagram Here]

Figure A.1: Complete ESP32 Sensor Node Wiring Schematic

APPENDIX B: ESP32 FIRMWARE CODE LISTINGS

The complete Arduino IDE firmware source code for the ESP32 sensor node is included below. The firmware implements sensor polling, MQ-7 heating cycle management, RFID debounce logic, JSON serialisation, and HTTP POST transmission.

[Insert Arduino IDE Firmware Source Code Here]

APPENDIX C: ENVIRONMENTAL DASHBOARD SCREENSHOTS

The following figures illustrate the React.js Environmental Dashboard interface, showing the real-time sensor status panel, time-series chart display, and alert history log.

[Insert Dashboard Screenshot – Real-Time Status Panel]

Figure C.1: Environmental Dashboard – Real-Time Sensor Panel

[Insert Dashboard Screenshot – Historical Chart View]

Figure C.2: Environmental Dashboard – Historical Data Chart

APPENDIX D: STUDENT MANAGEMENT SYSTEM SCREENSHOTS

The following figures illustrate the Student Management System interface, showing the student registry, attendance log view, and payment status management panel.

[Insert SMS Screenshot – Student Registry]

Figure D.1: Student Management System – Student Registry View

APPENDIX E: RAW SENSOR DATA SAMPLE

Table E.1 presents a sample of 20 consecutive sensor data records from Session 1 (Morning session, 8:00–10:00), representative of the full dataset structure stored in the MongoDB sensor_readings collection.

Table E.1: Sample Sensor Data Records – Session 1 (First 20 records)

Time	Temp (°C)	Humid (%)	CO (ppm)	Noise (dB)	RFID UID	Alert Level
08:00:05	27.2	65.3	6	44	A3F2B1C4	None
08:00:10	27.2	65.4	6	45	—	None
08:00:15	27.3	65.4	7	46	D1E4F290	None
08:00:20	27.3	65.5	7	44	—	None
08:00:25	27.4	65.6	8	47	B2C3A1D5	None
08:00:30	27.4	65.6	8	48	—	None
08:00:35	27.5	65.7	9	49	F0E1C2B3	None
08:00:40	27.5	65.8	9	50	—	Advisory
08:00:45	27.6	65.8	9	51	A0B1D2E3	Advisory
08:00:50	27.6	65.9	10	50	—	Advisory
08:00:55	27.7	66.0	10	52	C4D5F6A7	Advisory
08:01:00	27.7	66.0	11	51	—	Advisory
08:01:05	27.8	66.1	11	53	—	Advisory
08:01:10	27.8	66.2	12	52	E8F9A0B1	Advisory
08:01:15	27.9	66.2	12	54	—	Advisory
08:01:20	27.9	66.3	13	53	—	Advisory
08:01:25	28.0	66.3	13	55	B3C4E5F6	Advisory

08:01:30	28.0	66.4	14	54	—	Advisory
08:01:35	28.1	66.5	14	56	—	Advisory
08:01:40	28.1	66.5	15	55	D7E8F9A0	Advisory

APPENDIX F: Q-TABLE STATE-ACTION MAPPING

Table F.1 presents a representative subset of the Q-table after ten training episodes, showing Q-values for selected state indices. States are identified by their four-band tuple (Temperature Band × Humidity Band × CO Band × Noise Band), where 0=Low, 1=Normal, 2=High.

Table F.1: Q-Table Sample (Selected States After 10 Training Episodes)

Idx	State (T,H,CO,N)	Q(No Alert)	Q(Advisory)	Q(Critical)
0	(Low, Low, Low, Quiet)	0.82	0.12	-0.31
13	(Normal, Normal, Normal, Normal)	0.71	0.34	-0.18
26	(High, Normal, Normal, Normal)	0.21	0.68	0.15
40	(High, High, Normal, Normal)	0.08	0.72	0.28
53	(High, High, High, Normal)	-0.12	0.41	0.65
67	(High, High, High, Loud)	-0.34	0.18	0.71
80	(High, High, High, Loud)	-0.41	0.09	0.77

The Q-table demonstrates expected patterns: states with all parameters in Low or Normal bands favour the No Alert action (high Q-values for A0), while states with multiple High-band parameters increasingly favour Advisory (A1) and Critical (A2) actions. The progression from state 0 (all low) through state 80 (all high) shows a clear and interpretable shift in learned policy, confirming that the Q-Learning agent is acquiring meaningful environmental knowledge.